

Probability and Random Processes

Markov chains. Exercises

1. A ticket sales system utilises two computers: one operates actively, while the other remains idle and is designated to assume operations if the active computer fails. The probability that the active computer fails during a given day is $0 < p < 1/2$. If a failure occurs, the affected computer becomes operative again only on the second day following the failure.

Let (x, y) represent the state of the system, where x denotes the number of computers ready for operation and y indicates the number of days a computer has been under repair.

- (a) Model the system using a Markov chain. Classify all possible states and determine the corresponding transition probability matrix.
 - (b) Assuming both computers are operative today, identify the most probable system state two days later.
 - (c) Compute the stationary distribution. In the stationary regime, determine the probability that neither computer is operative on a given day when $p = 0.3$.
2. An elevator operates by moving up and down, stopping at each floor of the building. It has a maximum capacity of three passengers. At each floor, one of the following events can occur: either one passenger gets in, or one passenger gets off, or no one gets on or off. The probability of one passenger boarding the elevator is $1/4$ when the elevator is not full, and it is zero when the elevator is at full capacity. Conversely, the probability of one passenger getting off is $1/4$ when the elevator is not empty, and it is zero when the elevator is empty.

- (a) Determine the Markov chain, classify its states and compute its transition probability matrix.

If there are currently i passengers in the elevator,

- (b) What is the most probable number of passengers after two stops?
 - (c) In the long term, what is the average number of passengers?
 - (d) If T_i denotes the time (number of stops) that the elevator is full for the first time, what is the expected value of T_i ?
3. Consider a system consisting of four balls, two white and two black. Two balls are placed in urn A and the remaining two in urn B . The state of the system is defined by the number of white balls in urn A . The following process is repeatedly applied: a ball is selected at random from urn A and transferred to urn B , followed by the random selection of a ball from urn B to be transferred to urn A .
- (a) Determine the transition probability matrix $\mathbf{P}$ for the associated Markov chain.
 - (b) Identify the stationary distribution. Assess whether this stationary distribution also serves as a limiting distribution and provide justification. If applicable, compute $\lim_{n \rightarrow \infty} \mathbf{P}^n$.
 - (c) Suppose two balls are randomly selected from the four. Calculate the probabilities of drawing 0, 1, or 2 white balls. Relate these results to the findings from the previous question.
4. Consider a particle performing a random walk on a directed graph with vertex set $\{1, 2, 3, 4\}$ and adjacency matrix $\mathbf{A} = (a_{ij})$:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Here, $a_{ij} = 1$ if and only if there is a directed edge from node i to node j ; otherwise, $a_{ij} = 0$. If the particle is located at node i , it randomly selects one of its adjacent neighbors (with equal probability) and moves to that node.

- (a) Analyse the random walk using a Markov chain framework. Identify the type of Markov chain represented.
 - (b) Determine the stationary distribution. Assess whether it also serves as a limiting distribution and provide justification. For each node i , $1 \leq i \leq 4$, compute the expected number of steps required for the particle to return to i .
 - (c) Suppose the random walk begins at node i , where $3 \leq i \leq 4$. Calculate the expected number of steps until the particle first visits either node 1 or node 2. Solve this problem using two approaches: first-step analysis and by converting the Markov chain into an absorbing chain.
5. Consider two urns containing a total of four balls. At each step, one of the four balls is randomly selected and transferred from its current urn to the other urn. Define the states as the number of balls in the first urn.
- (a) Provide justification that the process constitutes a Markov chain and determine the corresponding transition probability matrix.
 - (b) Determine the stationary distribution. Assess whether this stationary distribution also serves as a limiting distribution and provide justification for your answer.
 - (c) Assume initially that the first urn contains all four balls. Calculate the expected number of movements required until it contains all four balls again.
 - (d) Now, suppose that each urn initially contains two balls. Calculate the expected number of movements required until all four balls are located in a single urn.
6. Analyse an absorbing Markov chain with state space $S = \{1, 2, 3, 4\}$, given the following transition probabilities.

$$p_{11} = 1, \quad p_{21} = \frac{1}{2}, \quad p_{22} = 0, \quad p_{23} = \frac{1}{4}, \quad p_{33} = 1, \quad p_{41} = \frac{1}{2}, \quad p_{42} = \frac{1}{2}.$$

- (a) Classify all states and present the transition probability matrix in canonical form.
 - (b) Calculate the fundamental matrix, then use it to determine the absorption probabilities and the expected times to absorption.
 - (c) Recompute these parameters using a first-step analysis.
7. Consider a binary sequence of length four, denoted as $x_1x_2x_3x_4$, where each x_i belongs to $\{0, 1\}$. At each step, one of the four bits x_i is selected uniformly at random, and its value is flipped with probability $1/2$. Let X_n represent the number of ones in the sequence after n steps.
- (a) Demonstrate that if $X_n = i$ for $0 \leq i \leq 4$, then the probability that $X_{n+1} = i + 1$ is $(4 - i)/8$. Determine the transition probability matrix for the Markov chain $\{X_n : n \geq 0\}$. Assess whether the chain is irreducible and aperiodic, providing justification for each conclusion.
 - (b) Find the stationary distribution and demonstrate that $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \binom{4}{k}/16$. Provide an explanation for this result.
 - (c) Assuming the initial sequence is 0000, apply first-step analysis to determine the expected number of steps required to obtain a sequence containing two ones and two zeros. Additionally, calculate the expected number of steps if the initial sequence is 1000 or 0010.
 - (d) Re-examine the previous question by modelling the process as an appropriate absorbing Markov chain.

8. Let X_n be a Markov chain with state set $S = \{0, 1, 2, 3\}$ and transition probabilities:

$$\begin{aligned} p_{11} &= p_{22} = p_{33} = \frac{1}{4}; \\ p_{10} &= p_{20} = p_{30} = \frac{1}{4}; \quad p_{01} = p_{02} = p_{03} = \frac{1}{3}; \\ p_{12} &= p_{13} = p_{21} = p_{23} = p_{31} = p_{32} = \frac{1}{4}. \end{aligned}$$

- (a) Draw the transition diagram, write the transition probability matrix $\mathbf{P}$, and find the stationary distribution.
 - (b) Let $T_0 = \min\{n \geq 1: X_n = 0\}$. Using first step analysis, compute the expected times $m_{i0} = \mathbb{E}(T_0 | X_0 = i)$ for each $i \in S$. Apply the fundamental matrix of the chain to verify the results obtained for each m_{i0} . Provide an explanation for the value of m_{00} .
9. Consider a game in which a coin with probability $\mathbb{P}(\text{heads}) = 2/3$ is tossed repeatedly. Denote the outcomes “heads” and “tails” by H and T, respectively. The game terminates upon the first occurrence of either the pattern HH or HTH, observed in two or three consecutive tosses, respectively.
- (a) Define an absorbing Markov chain with state space $\{\emptyset, \text{H}, \text{HT}, \text{HH}, \text{HTH}\}$ to model this game. Classify each state and provide the corresponding transition probability matrix.
 - (b) Calculate the expected number of coin tosses required for the game to conclude.
 - (c) Given that the most recent toss resulted in “heads”, determine the expected number of remaining tosses until the game concludes.
 - (d) Determine the probability that the game concludes with the pattern HH and the probability that it concludes with the pattern HTH.
10. A vending machine dispenses items for one euro and is equipped with a box, denoted B , with a capacity of $2k$ one-euro coins. Initially, box B contains k coins.
- Each customer independently pays with a one-euro coin with probability p (where $0 < p < 1$), depositing the coin into box B , or with a two-euro coin with probability $q = 1 - p$, depositing the coin into a separate box B' . In the latter case, the machine dispenses one euro from box B as change. When box B becomes empty, an employee refills it with k one-euro coins. When box B becomes full, the employee removes k one-euro coins.
- Let X_n denote the number of coins in box B after n individuals (including both customers and employees) have interacted with the machine.
- (a) Give the transition probabilities of the Markov chain $\{X_n: n \geq 0\}$ and classify its states.
 - (b) Let $k = 2$. Determine the stationary distribution. What is the expected number of individuals who have interacted with the machine until box B again contains two coins? Compute $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k)$ if this limit exists, and justify the answer.
 - (c) Let $p = 1/2$. Determine the expected number of customers served before the machine becomes empty or full for the first time.
- [Hint: Apply first-step analysis to verify that the service time, starting with i coins in B , is $i(2k - i)$.]